IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc Control Systems

GAME THEORY

Monday, 11 May 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : R.B. Vinter
Second Marker(s) : D. Angeli

© Imperial College London

Information for candidates:

No special instructions for candidates.

©Imperial College London

1. Consider a 2-player matrix game with pay-off matrix

$$S^A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \text{ and } S^B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

Player A (who chooses a row) and Player B (who chooses a column) both aim to minimize their pay-offs.

Assume that $0 < \alpha < 1$ and and $0 < \beta < 1$ where

$$\alpha = \frac{a_{21} - a_{11}}{a_{12} + a_{21} - a_{11} - a_{22}} \text{ and } \beta = \frac{b_{12} - b_{11}}{b_{12} + b_{21} - b_{11} - b_{22}}\,.$$

(a): Let $((1-\bar{\mu}, \bar{\mu}), (1-\bar{\nu}, \bar{\nu}))$ be a Nash equilibrium in mixed strategies for the two players. Assume the strategies are strictly mixed, in the sense that $0 < \bar{\mu} < 1$ and $0 < \bar{\nu} < 1$.

(i): Derive the Nash equilibrium conditions satisfied by $((1-\bar{\mu}, \bar{\mu})), (1-\bar{\nu}, \bar{\nu}))$. [5]

(ii): Hence show that $\bar{\mu} = \beta$. [5]

(b): Let $(1-\mu^*, \mu^*)$ be a safety strategy (in mixed strategies) for the Player A. Assume that the safety strategy is strictly mixed.

(i): Derive a relation of the form $\mu^* = \underset{\mu \in [0,1]}{\arg\min}\ \text{Max}\ \{\phi_1(\mu), \phi_2(\mu)\}$ for μ^* and determine the functions $\phi_1(.)$ and $\phi_2(.)$. [4]

(ii): Solve this relation to obtain a formula for μ^* in terms of the components of the pay-off matrices. [4]

(iii): If $S^A = -S^B$, i.e. the game is a zero sum game, show that $\bar{\mu} = \mu^*$, i.e. the first player's Nash equilibrium strategy coincides with the first player's safety strategy. [2]

©University of London 2007

2(a): N people must each decide whether to log in to the internet. Because of a bandwidth limit on the communication channel, no one will be connected if everybody, i.e. all N people, logs in. However everybody who logs in will be connected, if at least one person does not login.

The 'satisfaction' pay-off for Person i, $i = 1, \ldots, N$, which Person i aims to maximize, is quantified as follows:

Pay-Off	Scenario
$+1$	Person i logs in and at at least one person does not log in.
0	Person i does not log in
$-\gamma$	every person, including person i, logs in

Here, $\gamma > 0$ is a given constant. Person i is a participant in an N-player game, who must choose a mixed strategy:

'log in' with probability λ_i and 'don't log in' with probability $(1 - \lambda_i)$

for some $0 < \lambda_i < 1$. Assume that each person chooses to login in randomly, independently of the other people.

Derive the number $0 < \bar{\lambda} < 1$, such that the collection of mixed strategies for all of the players $(\lambda_1 = \bar{\lambda}, \ldots, \lambda_N = \bar{\lambda})$ is a Nash equilibrium. [8]

What is the probability that everyone logs in (and, consequently, no one is connected)? [2]

2(b): Consider the 3 stage zero sum game, in extended form, summarized in Fig. 2, involving players A and B. Players A and B aim to minimize and maximize the pay-off, respectively.

Stage 1: A chooses a strategy, left ('L') or right ('R').

Stage 2: B observes the strategy of A from Stage 1 and chooses a strategy, left ('L') or right ('R').

Stage 3: A chooses a strategy, left ('L') or right ('R'). (A remembers her strategy from Stage 1, but has not observed the strategy of B at Stage 2.)

(The information sets are indicated in the figure.)

Reduce the game to normal form. [7]

Identify any Nash equilibria, and state whether they are admissible. [2], [1]

©University of London 2007

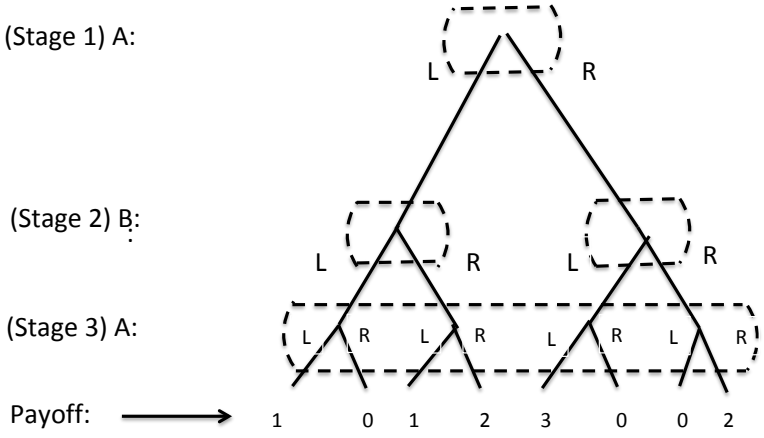


Figure 2: Extensive Form Game

©University of London 2007

3. $N(\geq 2)$ companies compete to sell an identical product. Let $x_i(> 0)$ be the level of production for Company i. The unit sale price for each company, $s(x_1, \ldots, x_N)$, is the same for each company and depends on the total production level of all the companies. It is given by the formula

$$s(x_1, \ldots, x_N) = (1 - \sum_{i=1}^{N} x_i)\,.$$

If Company i sells x_i items, the production cost $\phi(x_i)$ for Company i is

$$\phi(x_i) = c \times x_i$$

in which c, the unit production cost, is a constant $0 < c < 1$.

For each i, Company i aims to choose its level of production x_i to maximize Company i's profits: which are

$$\text{Profits of Company } i \;=\; x_i \times (1 - \sum_{i=1}^{N} x_i) - \phi(x_i)\,.$$

(a:) Find the number $y(N)$ (it will depend on N) such that $\{x_1 = y(N), \ldots, x_N = y(N)\}$ is an N-player Nash equilibrium with respect to the profits of each company and evaluate the profit for each company at equilibrium. [7]

Sketch the dependence of the unit sale price $P(N)$ (under Nash equilibrium conditions) on the number of companies N. [3]

(b:) Now assume that Company 1 acts as a leader and informs Companies $2, \ldots, N$ that Company 1's production level will be x_1.

Find the number $z(x_1)$ (it will depend on x_1) such that

$$\{x_2 = z(x_1), x_3 = z(x_1) \ldots, x_N = z(x_1)\}$$

is a Nash equilibrium for Companies $2, \ldots, N$, each of which seek to maximize its profits. [5]

Find the value of x_1 that maximizes the profits of Company 1. [5]

©Imperial College London

4(a): Consider the zero sum matrix game with pay-off matrix

$A\backslash B$		
	-1	$+\alpha$
	$+\alpha$	v

in which v and α are numbers such that $-1 \leq v \leq +1$ and $\alpha \geq 1$. Player A seeks to maximize the pay-off and the B player seeks to minimize the pay-off. Compute the saddlepoint equilibrium, in mixed strategies, for the two players. [6]

show that the value of the game (i.e. pay-off for saddlepoint equilibrium), is

$$\frac{\alpha + v}{1 + 2\alpha - v}\,.$$

[2]

4(b): Two players (Player A and Player B) compete in a zero sum dynamic game over the time horizon $t = 0, 1, \ldots, N$. At any time t, each player can choose a mixture of strategy S_1 and strategy S_2.

Let v_t be the value of the game at time t, i.e. the Nash equilibrium pay-off of the game over the time horizon $t, t+1, \ldots, N$.

The pay-off matrix at time t is related to v_{t+1} (the value of the game at the time t_{t+1}) according to the tableau:

$A\backslash B$	S_1	S_2
S_1	-1	$+1$
S_2	$+1$	v_{t+1}

Using the results of 4(a) above, or otherwise,

(i): Derive the 'backwards' recursive formula for the value v_t, $t = 0, \ldots, N$ of the game, i.e. find the function $\phi(v)$ such that

$$v_{t-1} = \phi(v_t), \qquad \text{for } t = N, N-1, \ldots, 0\,. \tag{4}$$

Hence show that, for the boundary condition $v_N = 0$, the value function is given by the formula

$$v_{N-m} = \frac{m}{m+2}, \qquad \text{for } \; m = 0, 1, \ldots. \tag{4}$$

(You need only check this formula for the first few values of m, $m = 0, 1, 2, \ldots$)

(ii): Obtain a formula for the saddlepoint mixed strategies of the two players $((1-\mu_t, \mu_t)$, $((1-\nu_t, \nu_t)$, respectively, at time t. [4]

(*Interpretation of the game in part (b): The two players are two companies competing for a contract (S_1 ='bid', S_2= 'not bid'). If both companies choose S_2 ('not bid'), the competition is delayed until following year, but not beyond N years.*)

©Imperial College London

5. Consider the non-linear control system described by the state equation

$$x_{t+1} = \frac{x_t + u_t(1+x_t)}{1 - u_t(1+x_t)} + u_t \qquad \text{for } t = 0, \ldots, N-1 \tag{1}$$

and define the function

$$\phi(x_0, \{u_t\}) := \sum_{t=0}^{N-1} u_t^2 + \left| \frac{x_N}{1+x_N} \right|^2 ,$$

in which $x_1, \ldots, x_N$ are obtained from the control history $\{u_t\}$ and the initial state x_0 by solving the state equation (1).

Consider the optimal control problem with state equation (1), cost function $J(x_0, \{u_t\}) := \phi(x_0, \{u_t\})$ and fixed initial state x_0.

Formulate the Bellman equation for this problem. [4]

Determine the control history, in state feedback form, that minimizes the cost $J(x_0, \{u_t\}) = \phi(x_0, \{u_t\})$. [12]

Determine the minimum cost [4]

(*Hint: seek a solution $V_t(x)$ to the Bellman equation having the structure*

$$V_t(x) = p_t \left| \frac{x}{1+x} \right|^2 \qquad \textit{for } t = 0, \ldots, N ,$$

for some constants $p_0, \ldots, p_N$.)

©Imperial College London